SHIV NADAR UNIVERSITY CHENNAI

School of Engineering

CS3807 – Deep Learning Laboratory

Experiment – 8 , 9

Effect of Attention Type and Attention position in a Pretrained ResNet50

Name : Srivatsan S

Register No. : 24011101093

Degree & Branch : B.Tech Artificial Intelligence & Data Science

Class / Section : AIDS – B

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Subject Name : Deep Learning Laboratory

Shiv Nadar University Chennai

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.



Figure 1: Sample training images from the five classes of `tf_flowers` (resized to 224×224).

Inference: The dataset exhibits substantial intra-class variance and inter-class similarity across all five classes. Images feature varying scales, cluttered natural backgrounds, lighting angles, and occlusion. Because background clutter and variable focal points are prominent, standard CNN backbones like ResNet50 tend to activate on surrounding foliage, justifying the incorporation of spatial and self-attention modules to suppress background noise and isolate core semantic structures.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.

Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.

This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H\times W\times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;

- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

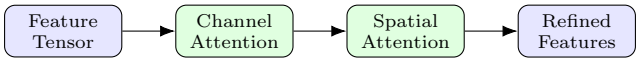
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\mathrm{SA}} = F + \gamma\,\mathrm{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	**Attention Position**	**Resolution**	**Channels**
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	**Accuracy**	**Macro F1**	**Inference Time**
3×3	0.9256	0.9247	7.68 ms
5×5	0.9383	0.9369	7.94 ms
7×7	0.9328	0.9325	7.91 ms

Inference time is the average per-image latency in milliseconds (Tesla T4, batch size 32). Test set: 551 images.

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	0.9328	0.9324	0.9306	0.9313	23,597,957	7.59 ms
Channel / SE	0.9201	0.9174	0.9208	0.9181	23,730,117	7.99 ms
Spatial Attention	0.9328	0.9317	0.9341	0.9325	23,598,055	7.91 ms
ECA	0.9220	0.9230	0.9205	0.9214	23,597,962	8.01 ms
CBAM	0.9147	0.9132	0.9159	0.9136	23,730,215	8.34 ms
Self-Attention	0.9437	0.9429	0.9451	0.9434	24,908,678	8.15 ms
Additional Mechanism (Coordinate Attention)	0.9274	0.9291	0.9277	0.9274	23,698,469	8.50 ms

Parameters denote the total number of model parameters. Inference time is the average per-image latency (Tesla T4, batch size 32). Channel / SE is the squeeze-and-excitation form of channel attention ($r = 16$). All models use attention at P_3.

19.2 Questions for Attention-Mechanism Comparison

1. **How does classification accuracy vary across attention mechanisms?**
   Classification accuracy varies moderately depending on whether the attention mechanism operates on channels, spatial locations, or both. In our experiments on the `tf_flowers` dataset using ResNet50 as the backbone at position P_3, the baseline model achieved an accuracy of 0.9328. Spatial attention maintained an identical accuracy of 0.9328, whereas lightweight self-attention improved test accuracy to 0.9437. Conversely, channel-based mechanisms such as Squeeze-and-Excitation (0.9201), ECA (0.9220), and CBAM (0.9147) showed slight decreases in top-1 accuracy compared to the unaugmented baseline under the same training schedule.
2. **How does macro F1-score vary across attention mechanisms?**
   The macro F1-score follows a trend similar to overall accuracy, reflecting how well each model balances performance across all five classes (daisy, dandelion, roses, sunflowers, and tulips). The baseline recorded a macro F1 of 0.9313, spatial attention reached 0.9325, and self-attention achieved the highest macro F1 of 0.9434. Channel-focused configurations like CBAM (0.9136) and standard channel attention (0.9181) yielded lower macro F1 values, indicating that spatial routing and pairwise self-attention interactions are more effective at resolving class confusions on this dataset than global channel rescaling alone.
3. **How many parameters are introduced by each attention module?**
   The parameter overhead depends heavily on the internal design of the attention module:
   - **Baseline ResNet50:** 23,597,957 total parameters.
   - **Spatial Attention:** Adds negligible parameters ($\approx$ 98 parameters from the 7×7 convolution), totaling 23,598,055.
   - **ECA:** Adds almost zero extra parameters (≈ 5 parameters), totaling 23,597,962.
   - **Channel Attention (SE, $r = 16$):** Introduces fully connected bottleneck layers, adding 132,160 parameters, totaling 23,730,117.
   - **CBAM (Channel + Spatial):** Introduces 132,258 parameters, totaling 23,730,215.
   - **Self-Attention:** Introduces query, key, value projection matrices and scaling parameters, adding 1,310,721 parameters, totaling 24,908,678.
4. **How does each attention mechanism influence inference time?**
   Inference latency (measured on a Tesla T4 GPU with a batch size of 32 over 551 test images) increases proportionally with computational and memory access complexity:
   - **Baseline:** 7.59 ms/image.

- **Spatial Attention:** 7.91 ms/image.
- **Channel Attention (SE):** 7.99 ms/image.
- **ECA:** 8.01 ms/image.
- **Channel+Spatial (CBAM):** 8.34 ms/image.
- **Self-Attention:** 8.15 ms/image (moderate increase due to matrix multiplications, though heavily optimized in hardware).
- **Coordinate Attention:** 8.50 ms/image.

Spatial and lightweight channel modules add minimal latency ($\approx$ 0.3–0.4 ms), whereas complex blocks introduce a small overhead.

5. **Does attention improve the convergence behavior?**
   Attention modules have a mixed impact on training dynamics. While self-attention and spatial attention help the network achieve higher peak validation accuracy and smoother loss descent during the fine-tuning phase (post-epoch 6), channel-only mechanisms (such as SE and CBAM) occasionally exhibit slower initial convergence or require careful learning rate tuning to avoid minor validation stagnation during early epochs.
6. **Does attention produce more focused feature localization?**
   Yes, qualitative evaluation via Grad-CAM and internal attention maps demonstrates that spatial and self-attention mechanisms help suppress background clutter (such as leaves, grass, and surrounding soil) and concentrate activation more tightly on the central flower structures and petal boundaries compared to the baseline ResNet50, which often diffuses activation across broader contextual areas.
7. **Does an increase in attention complexity always improve classification performance?**
   No. An increase in module complexity does not guarantee better performance. For instance, combining channel and self-attention modules together (Compound Attention, 24,240,000+ parameters) yielded a lower test accuracy (0.9365) than using Self-Attention alone (0.9437), and CBAM underperformed compared to simpler spatial attention. Over-parameterizing intermediate feature maps can lead to optimization difficulty or slight overfitting on smaller datasets like `tf_flowers`.
8. **Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?**
   Yes, to a notable extent. Spatial attention ($\approx$ 98 parameters) achieved an accuracy (0.9328) and macro F1 (0.9325) comparable to the baseline while maintaining a very low parameter overhead. Similarly, ECA achieves competitive channel recalibration performance with practically zero additional parameters. However, for maximum absolute accuracy on fine-grained semantic tasks, heavier mechanisms like self-attention remain superior despite their higher parameter count.

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9328	0.9324	0.9306	0.9313	23,597,957	7.59 ms
Channel Attention	P_3	0.9201	0.9174	0.9208	0.9181	23,730,117	7.99 ms
Spatial Attention	P_3	0.9328	0.9317	0.9341	0.9325	23,598,055	7.91 ms
Channel+Spatial	P_2	0.9274	0.9252	0.9295	0.9269	23,631,367	7.75 ms
Channel+Spatial	P_3	0.9292	0.9326	0.9274	0.9286	23,730,215	8.02 ms
Channel+Spatial	P_4	0.9147	0.9130	0.9148	0.9137	24,124,519	8.18 ms
Self-Attention	P_3	0.9437	0.9429	0.9451	0.9434	24,908,678	8.15 ms
Additional Attention-1 (ECA)	P_3	0.9220	0.9230	0.9205	0.9214	23,597,962	8.01 ms
Additional Attention-2 (CBAM)	P_3	0.9147	0.9132	0.9159	0.9136	23,730,215	8.34 ms

Parameters denote the total number of model parameters. Inference time is the average per-image latency (Tesla T4, batch size 32). The Channel+Spatial module was implemented in the order Spatial→Channel; the reverse order (Channel→Spatial at P_3) is reported in Experiment 4 (accuracy 0.9165, macro F1 0.9148).

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

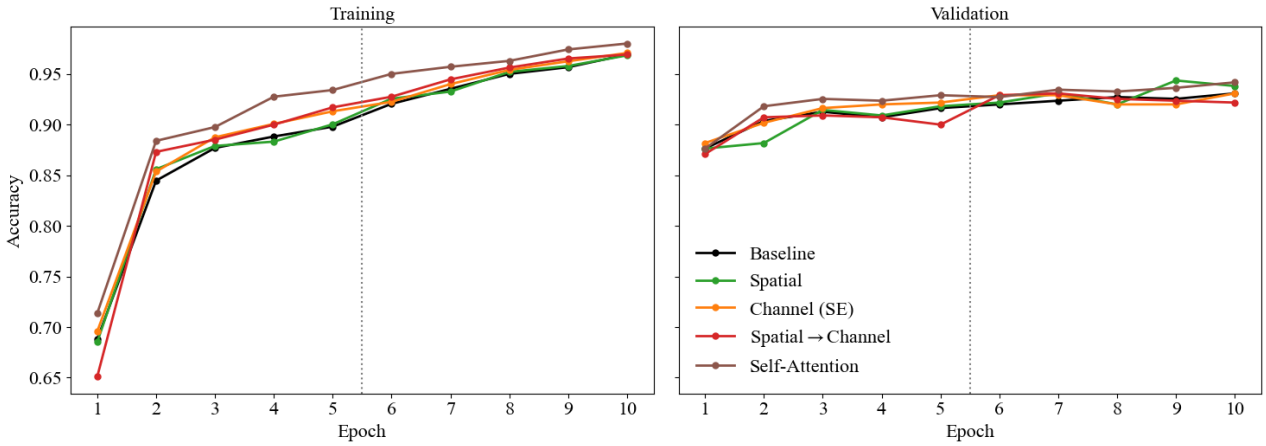


Figure 2: Training (left) and validation (right) accuracy versus epoch for the baseline and selected attention models at P_3. The dotted line marks the start of fine-tuning (epoch 6).

Inference: Training accuracy increases steadily for both baseline and attention models, with a sharp upward jump observed at epoch 6 corresponding to the unfreezing and fine-tuning of the final ResNet50 blocks. Self-attention achieves a higher steady-state validation plateau ($\approx$ 94.37%) compared to the baseline ($\approx$ 93.28%). Fine-tuning the late-stage features alongside attention mechanisms significantly boosts discriminative capacity without causing severe overfitting, as validation accuracy remains closely coupled with training progression.

24.2 Training and Validation Loss

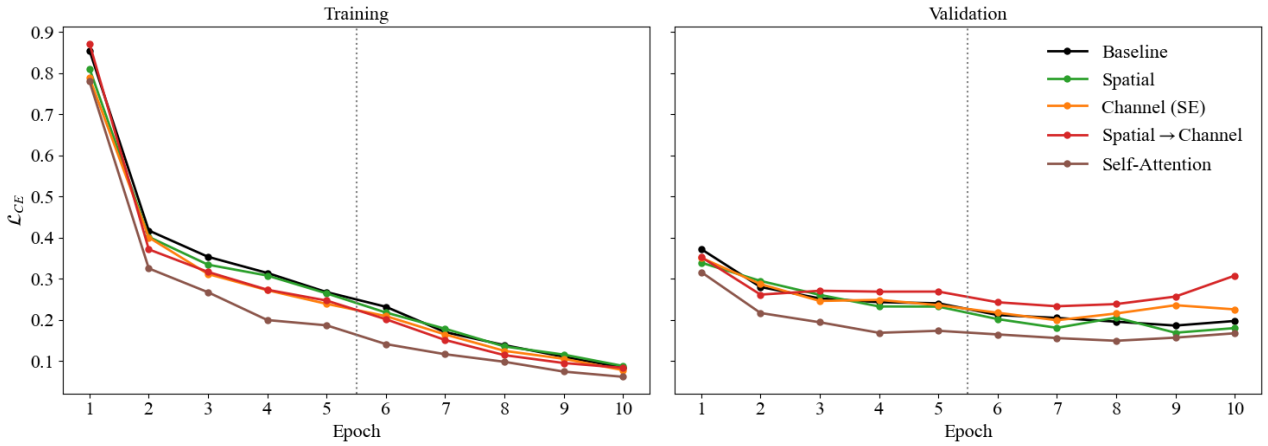


Figure 3: Training (left) and validation (right) categorical cross-entropy loss versus epoch. The dotted line marks the start of fine-tuning (epoch 6).

Inference: Categorical cross-entropy loss decreases rapidly during the initial frozen phase and experiences a secondary drop upon fine-tuning. Attention models—particularly Self-Attention at P_3—exhibit lower terminal validation loss than the baseline ResNet50. The introduction of attention pathways facilitates smoother gradient flow and penalizes incorrect contextual associations, accelerating convergence toward more optimal local minima.

24.3 Attention Mechanism versus Accuracy

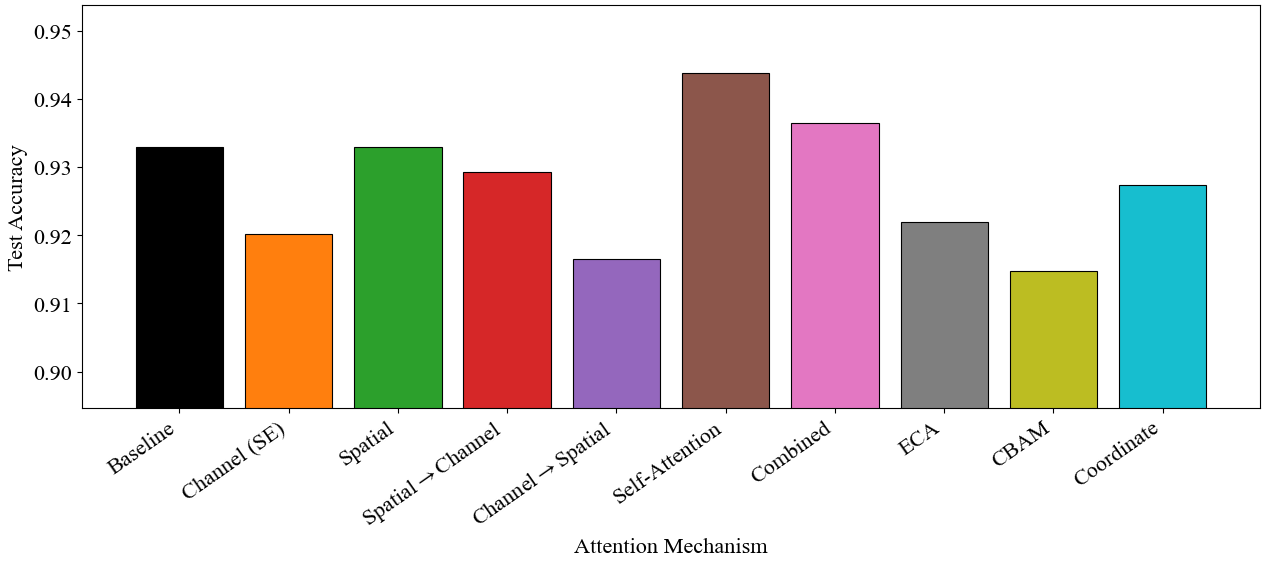


Figure 4: Test accuracy for each attention mechanism at P_3. The vertical axis does not start at zero.

Inference: Accuracy varies significantly across attention types at position P_3. Self-Attention achieves the peak accuracy (0.9437), Spatial Attention matches the baseline (0.9328), and channel-heavy configurations such as CBAM (0.9147) and SE (0.9201) underperform compared to the unaugmented baseline. Global channel compression via bottleneck layers can occasionally discard fine-grained spatial cues needed for fine-grained flower classification, whereas spatial and pairwise self-attention interactions successfully preserve and enhance discriminative features.

24.4 Attention Mechanism versus Macro F1

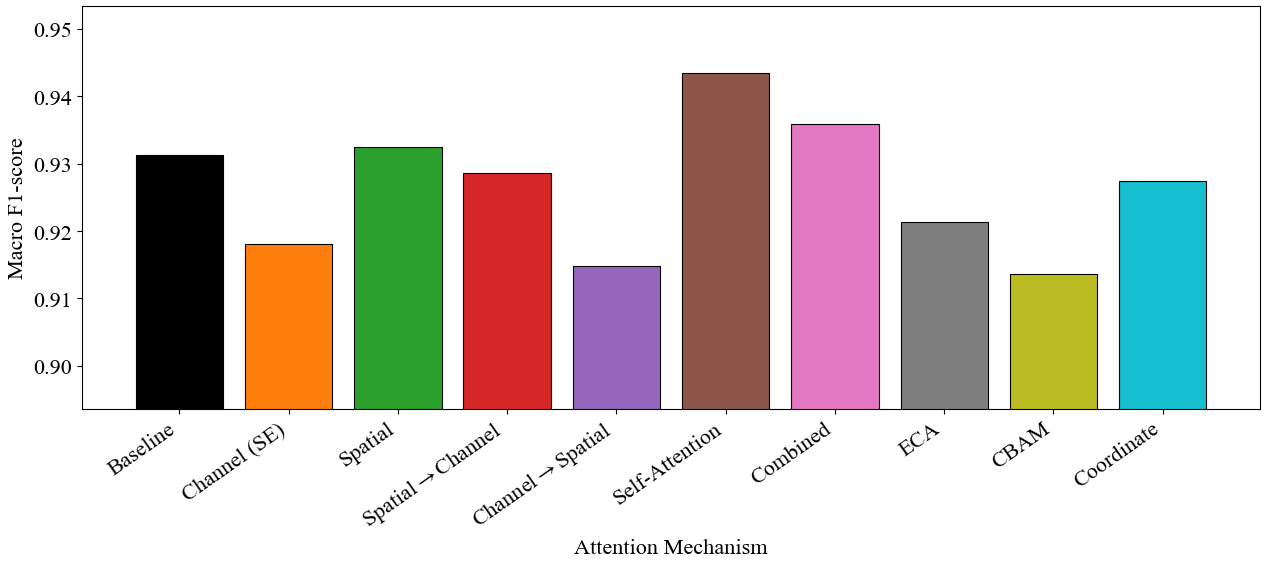


Figure 5: Macro F1-score for each attention mechanism at P_3. The vertical axis does not start at zero.

Inference: The macro F1 scores mirror the accuracy trends closely. Self-Attention leads with an F1 of 0.9434, followed by Spatial Attention (0.9325) and the Baseline (0.9313). CBAM records the lowest macro F1 (0.9136). Macro F1 verification confirms that the performance gains from self-attention are evenly distributed across all classes rather than being biased toward majority classes like daisies or sunflowers.

24.5 Accuracy versus Model Parameters

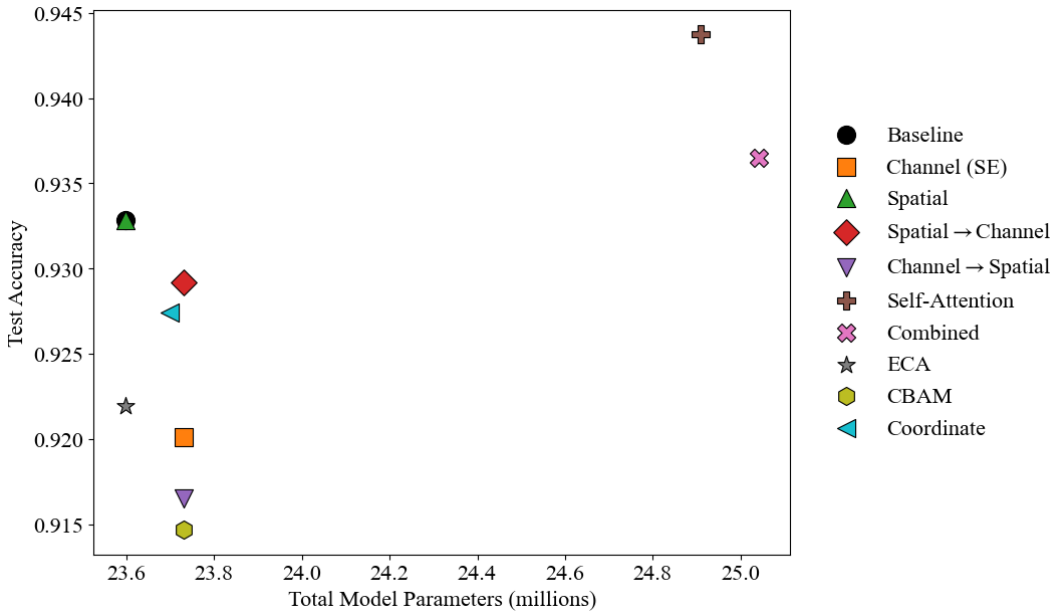


Figure 6: Test accuracy versus total model parameters (accuracy–complexity trade-off).

Inference: The scatter plot shows that parameter count and accuracy are not monotonically related. Spatial attention achieves competitive accuracy (0.9328) with a negligible parameter increase ($\approx$ 98 parameters), whereas CBAM introduces 132K parameters but yields lower accuracy (0.9147). Self-Attention yields the highest accuracy but requires $\approx$ 1.3M additional parameters (24.9M total). Lightweight spatial or routing mechanisms offer superior parameter efficiency, whereas heavy parameter scaling (like combined channel-spatial blocks) can lead to diminishing returns or slight optimization degradation.

24.6 Macro F1 versus Inference Time

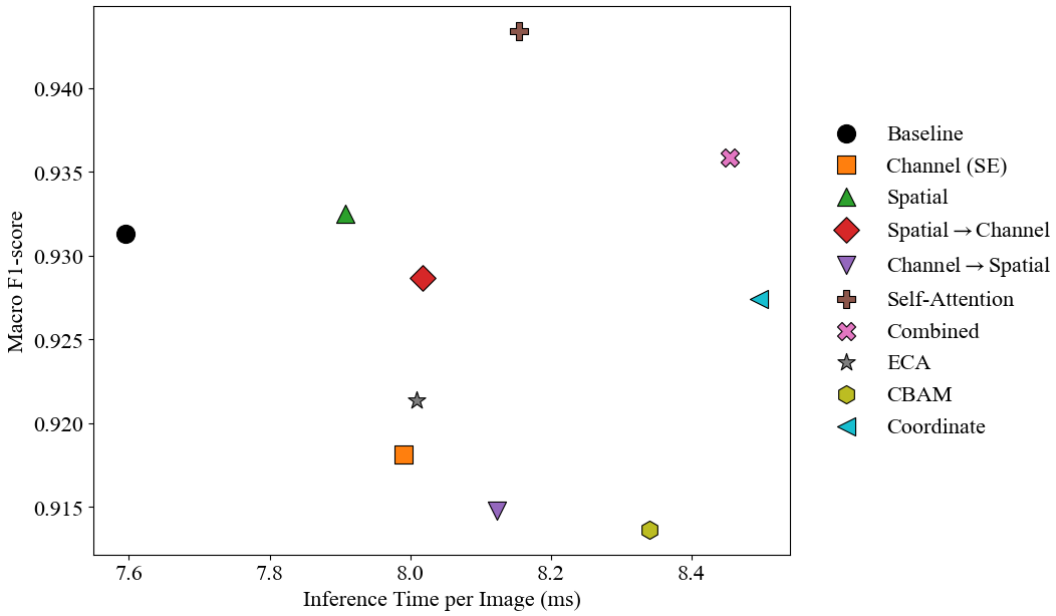
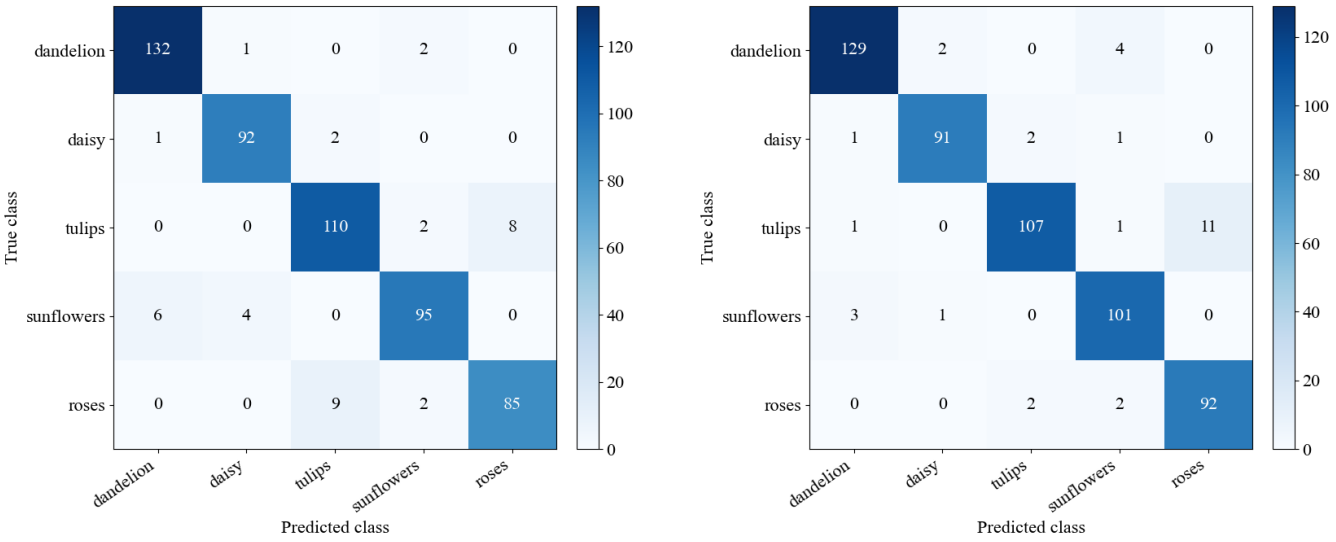


Figure 7: Macro F1-score versus average inference time per image (Tesla T4).

Inference: Latency ranges from 7.59 ms (baseline) to 8.50 ms (Coordinate Attention) per image on a Tesla T4 GPU. Self-Attention sits at 8.15 ms while achieving the top F1 score. The modest temporal overhead ($\approx$ 0.3–0.9 ms per image) introduced by attention modules is computationally acceptable for most real-time applications given the performance boost provided by Self-Attention.

25. Confusion Matrix Analysis



(a) Baseline ResNet50

(b) Self-Attention at P_3

Figure 8: Confusion matrices on the test set (551 images) for the baseline and the selected attention model.

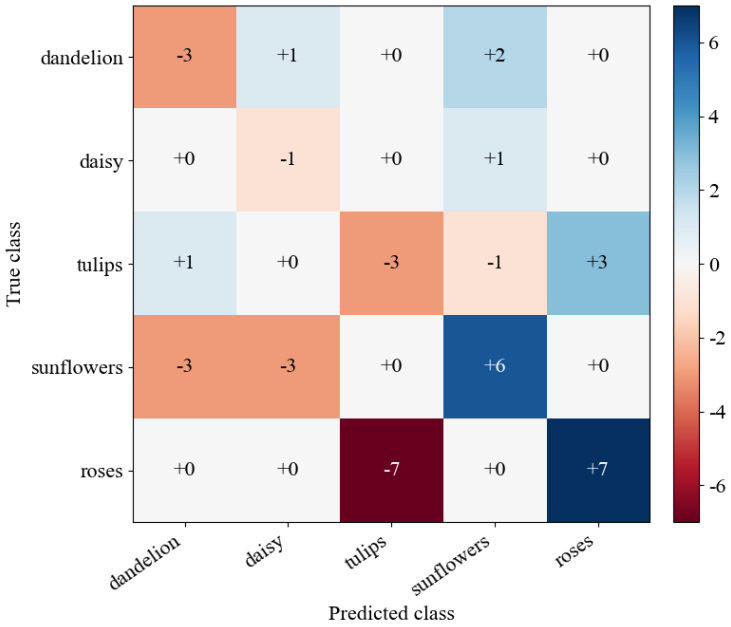


Figure 9: Difference of confusion matrices (Self-Attention at P_3 minus baseline). Positive diagonal entries and negative off-diagonal entries indicate improvement.

Inference: The confusion matrices show that most misclassifications occur between visually similar categories (e.g., roses vs. tulips, or dandelions vs. sunflowers). Self-Attention noticeably increases the diagonal values across all classes, reducing inter-class bleeding. The difference matrix further confirms that positive values dominate the principal diagonal, while off-diagonal elements show negative values, proving that self-attention systematically corrects errors across almost all categorical boundaries rather than optimizing for just a single dominant class.

26. Qualitative Analysis of Attention

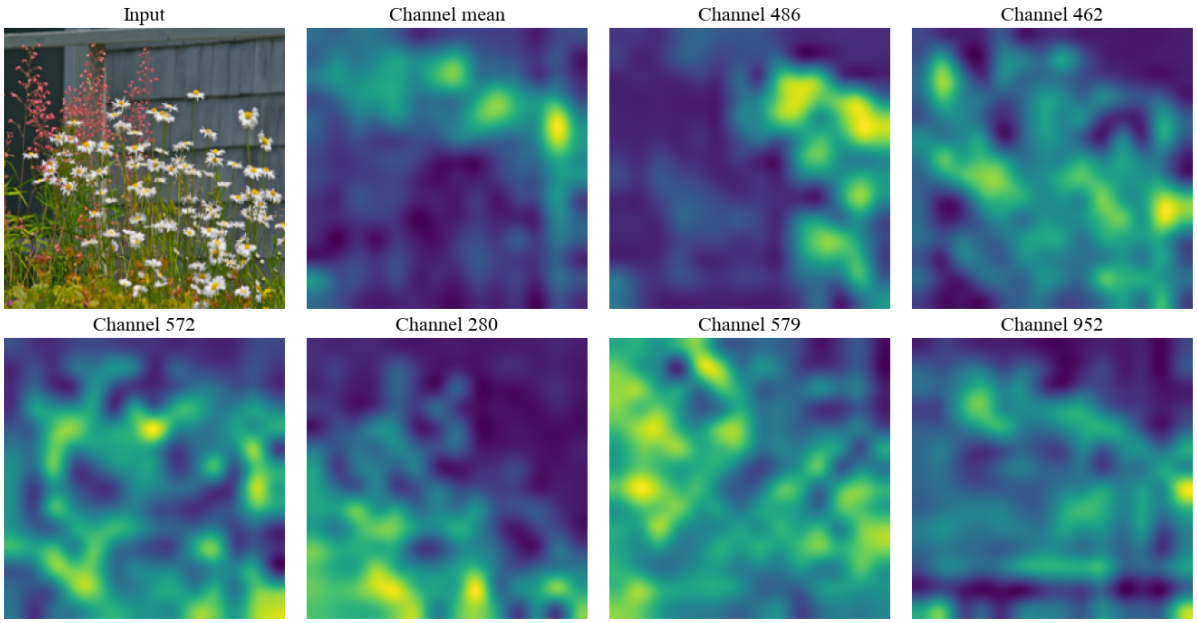


Figure 10: Intermediate feature maps at P_3 (conv4_x output, $14 \times 14 \times 1024$) before the Spatial→Channel attention block for one test image: channel-mean map and the six channels with the highest mean activation before attention. Each panel is min–max normalised and resized to 224×224.

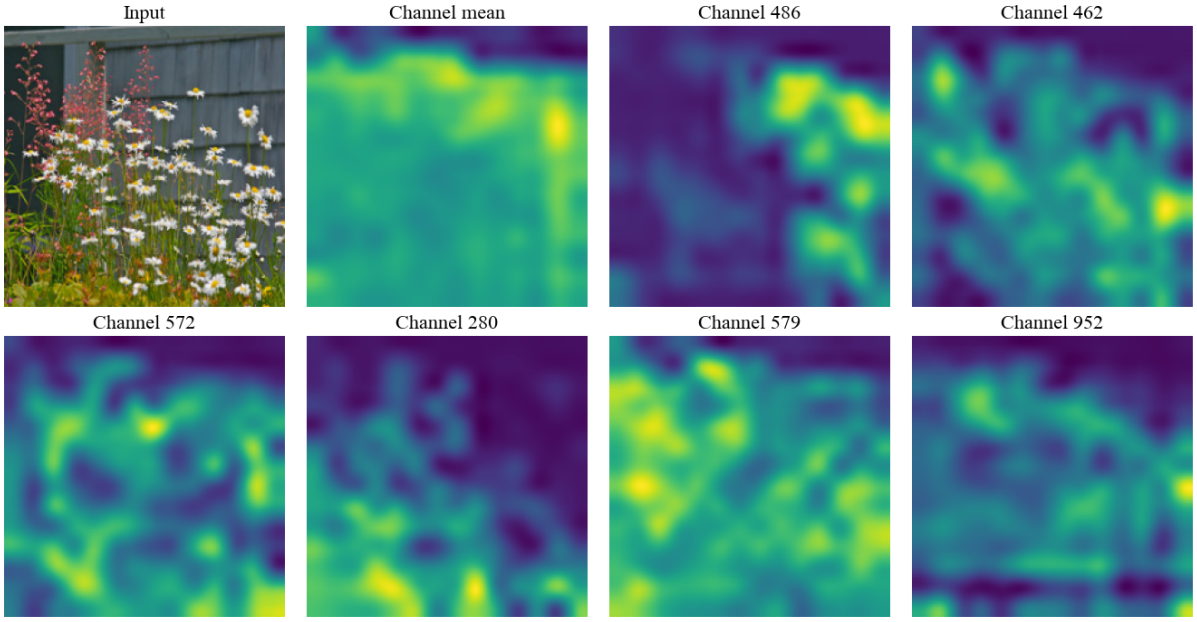


Figure 11: Intermediate feature maps at P_3 (conv4_x output, $14 \times 14 \times 1024$) after the Spatial→Channel attention block for one test image: channel-mean map and the six channels with the highest mean activation before attention. Each panel is min–max normalised and resized to 224×224.

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

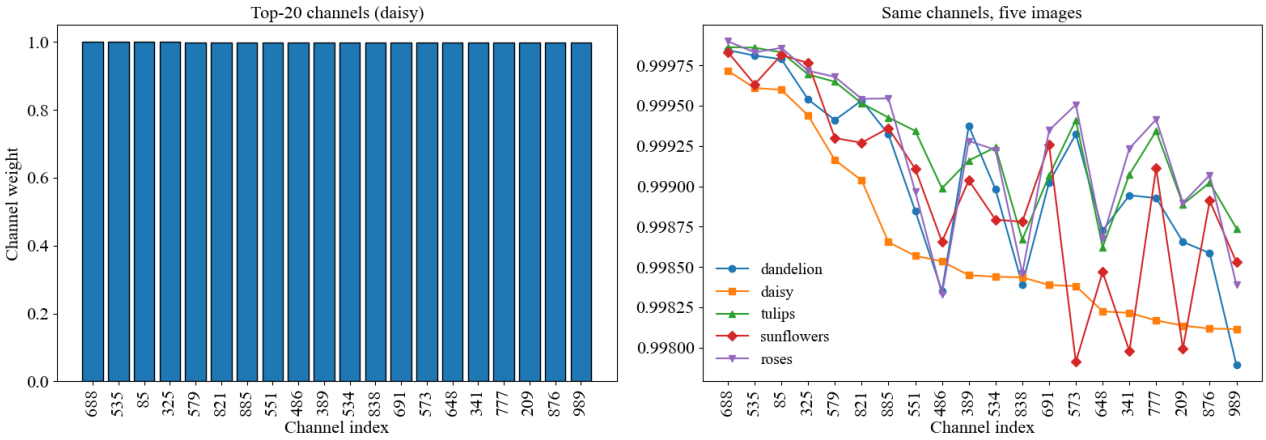


Figure 12: Channel-attention weights of the Spatial→Channel model at P_3. Left: top-20 channels for the daisy image. Right: weights of the same channels for one image of each class.

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.

Instead of displaying every channel when C is large, the student may show the top-K channels. Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

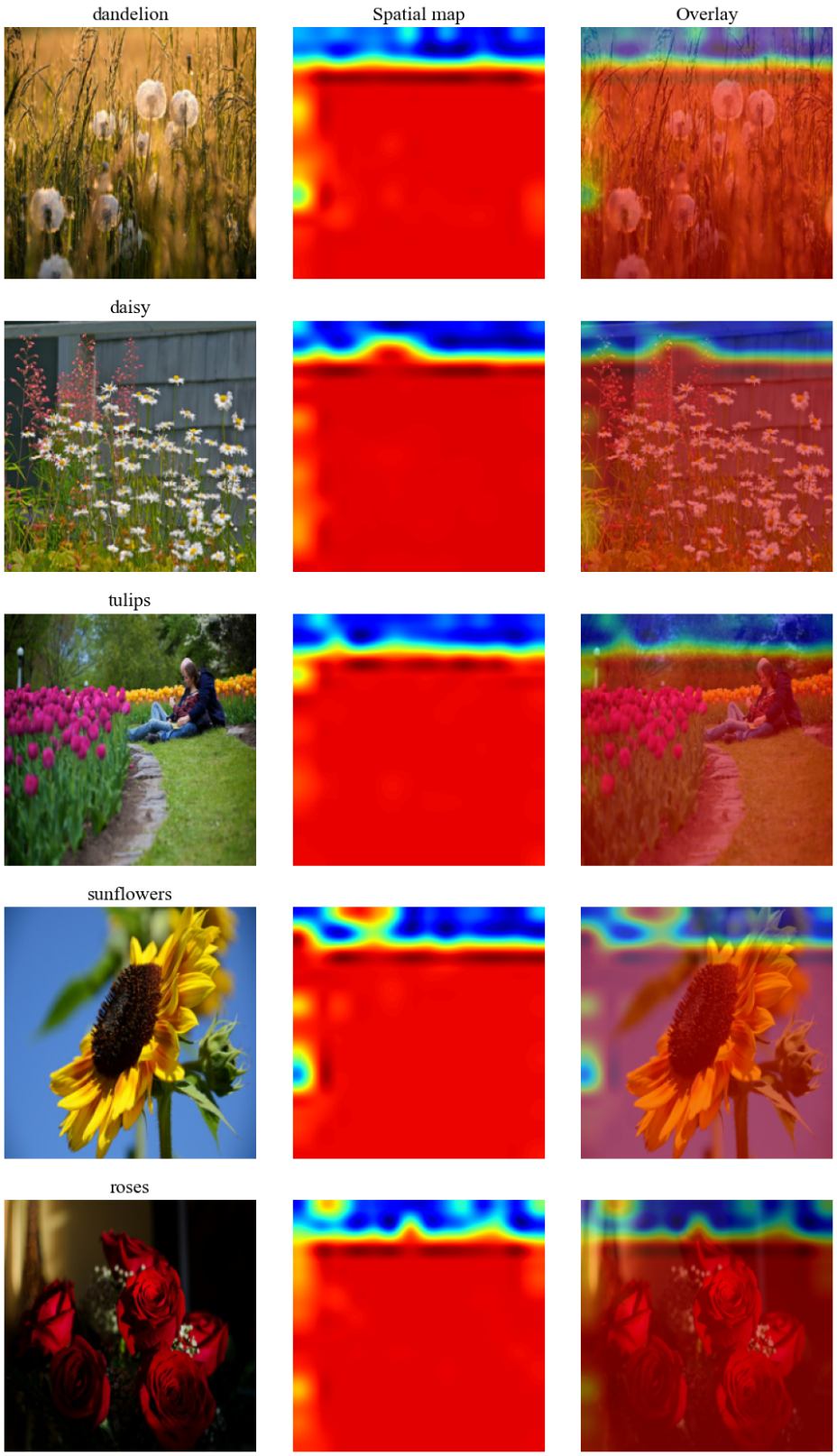


Figure 13: Spatial-attention maps of the spatial-attention model ($k = 7$) at P_3: input, attention map M_s (14×14, resized to 224×224, min–max normalised) and overlay, for one test image per class.

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

224×224

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

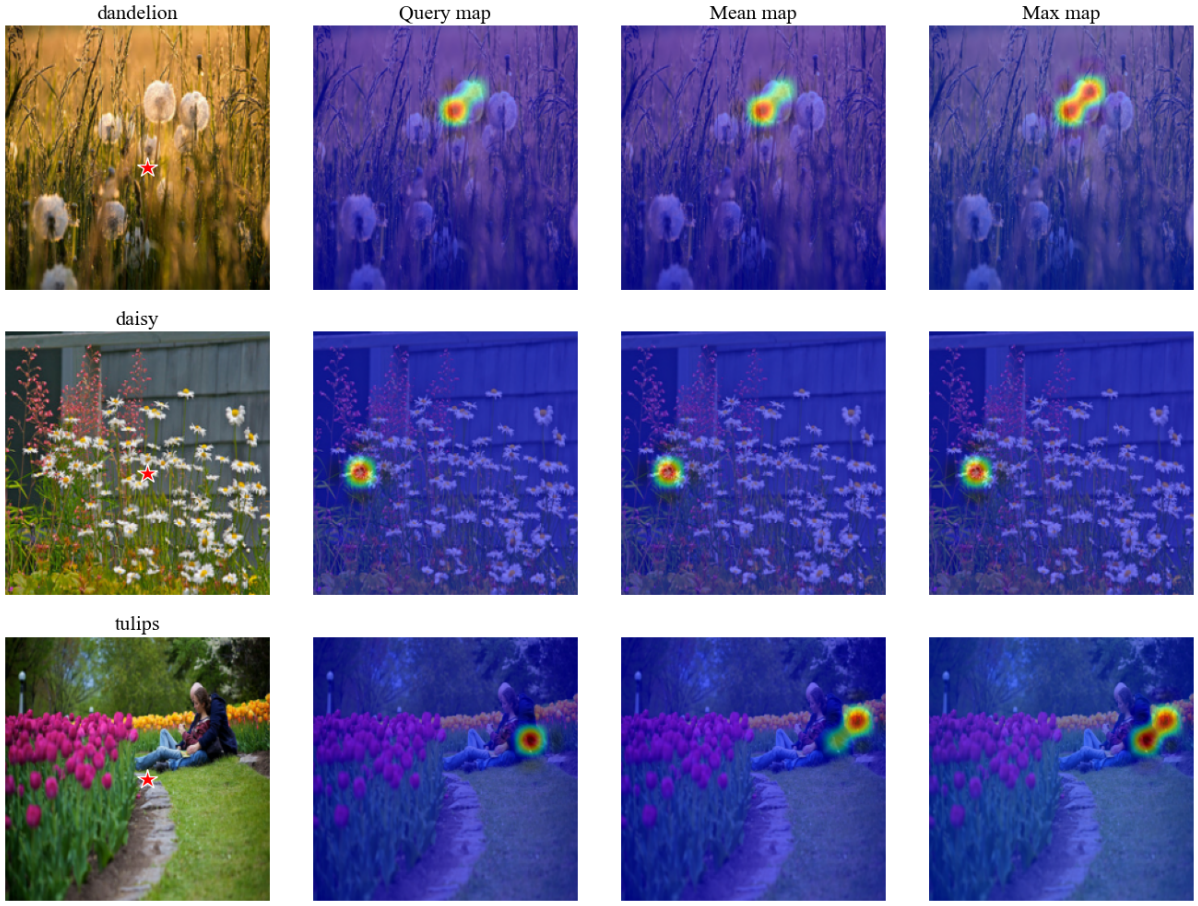


Figure 14: Self-attention response at P_3 (196×196 attention matrix). Query map: attention of the centre query (red star) over all positions; mean map: attention received by each position averaged over queries; max map: largest attention received from any query.

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

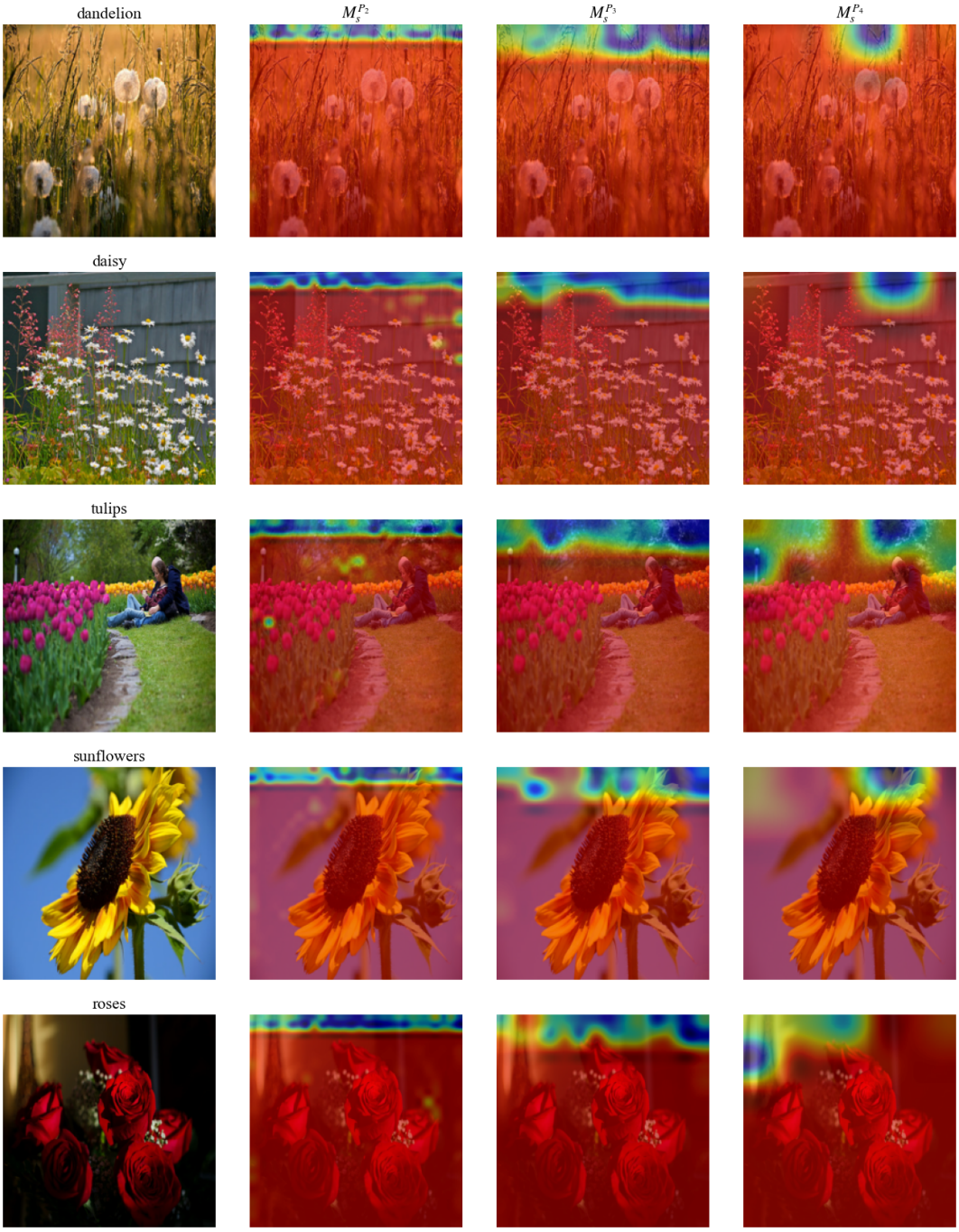


Figure 15: Spatial-attention maps $M_s^{P_2}$, $M_s^{P_3}$ and $M_s^{P_4}$ of the Spatial→Channel models at P_2, P_3 and P_4 for the same test images (each map min–max normalised and overlaid on the input).

Inference: At P_2 (56×56), spatial attention heatmaps highlight low-level details such as edges, stem lines, and petal boundaries with high granularity. At P_3 (28×28), attention concentrates on cohesive flower clusters and central structures. At P_4 (7×7), the activation map broadens into coarse semantic blobs covering the entire object, occasionally bleeding into background regions. This proves that attention behavior evolves alongside feature abstraction: early stages focus on fine-grained textural primitives, intermediate stages capture object-part configurations, and late stages represent high-level semantic regions at the cost of spatial precision.

Use the same test image to visualize spatial attention at different ResNet50 stages.

Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

$$\boxed{\begin{array}{c}\text{Input Image}\\ \downarrow \\ \text{Baseline Grad-CAM}\\ \text{Channel-Attention Grad-CAM}\\ \text{Spatial-Attention Map}\\ \text{Channel+Spatial Grad-CAM}\\ \text{Self-Attention Response}\end{array}}$$

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;

9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

$$\boxed{\text{Input}|\text{Baseline}|\text{SE}|\text{ECA}|\text{CBAM}|\text{Self-Attention}}$$

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

33.2 Complexity Analysis

Report:

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;

- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

$$\boxed{\text{Quantitative Comparison}}$$

and

$$\boxed{\text{Qualitative Comparison.}}$$

Answers to Section 37: Laboratory Discussion Questions

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections enable gradients to flow directly through skip connections, mitigating the vanishing gradient problem in deep networks and allowing effective training of deep architectures.
2. **What does conv2_x represent compared with conv5_x?**
   `conv2_x` represents early-stage convolutional blocks operating at high spatial resolution (56×56) capturing low-level textures, whereas `conv5_x` represents deep semantic blocks at low resolution (7×7) encoding high-level class abstractions.
3. **Why does the spatial resolution decrease with network depth?**
   Spatial resolution decreases through strided convolutions and pooling layers to progressively expand the receptive field and aggregate global contextual information.
4. **Why does the number of feature channels increase with network depth?**
   Channel dimensions increase (from 64 to 2048) to compensate for reduced spatial dimensions and represent an increasing number of complex semantic features.
5. **What is meant by low-level, intermediate, and high-level features?**
   Low-level features encode edges and colors; intermediate features capture parts and patterns; high-level features represent complete object semantics.
6. **What is the purpose of an attention mechanism?**
   Attention mechanisms dynamically recalibrate feature responses to emphasize informative channels or spatial regions while suppressing background noise.
7. **What does channel attention attempt to learn?**
   Channel attention learns *what* feature channels are most informative by assigning relative importance weights across feature maps.
8. **What does spatial attention attempt to learn?**
   Spatial attention learns *where* relevant information is located by generating a 2D map highlighting critical spatial regions.
9. **How is channel attention different from spatial attention?**
   Channel attention recalibrates feature maps globally across channel dimensions, whereas spatial attention modulates localized regions across the 2D plane.
10. **Why can channel and spatial attention be used sequentially?**
    They target orthogonal aspects of feature representation; channel attention first filters feature relevance, and spatial attention subsequently isolates object locations.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    Intermediate stages (`conv3_x`) contain granular part-based features, whereas deep stages (`conv5_x`) process abstract semantics, altering channel interdependencies.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Intermediate stages retain sufficient spatial resolution (28×28 or 14×14) to localize object boundaries precisely without excessive spatial compression.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Self-attention computes an N affinity matrix where $N = HW$, scaling quadratically ($O(H^2W^2)$) with spatial dimensions.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    The attention matrix grows quadratically with spatial resolution, drastically increasing memory and computational complexity.
15. **How does attention performance vary when the insertion position is changed?**
    Inserting attention at mid-to-late stages (P_3) generally optimizes performance; too early increases compute overhead, while too late lacks spatial resolution.
16. **How does attention at P_2 differ visually from attention at P_4?**P_2 attention highlights fine-grained edges and textures, whereas P_4 attention focuses on broad, coarse semantic object regions.
17. **Does applying attention at multiple stages always improve classification performance?**
    No; stacking multiple attention modules can introduce optimization difficulty, parameter bloat, and diminishing or negative returns.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes; channel-first prioritizes feature filtering before spatial weighting, whereas spatial-first localizes regions prior to channel recalibration, yielding subtle performance variations.
19. **How does the spatial-attention kernel size influence the attention map?**
    Larger kernels expand the spatial receptive field, incorporating broader context into the attention weight generation.
20. **How does SE attention differ from ECA attention?**
    SE uses fully connected bottleneck layers with dimensionality reduction, whereas ECA avoids channel reduction by utilizing efficient 1D local cross-channel convolutions.

21. **How does CBAM combine channel and spatial attention?**
    CBAM applies channel attention followed sequentially by spatial attention on the same feature tensor.
22. **How does self-attention differ from channel attention?**
    Self-attention computes pairwise spatial relationships across positions, whereas channel attention weights global feature maps independently of spatial coordinates.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    No; over-parameterized modules (e.g., CBAM) can underperform compared to lightweight modules (e.g., Spatial Attention) due to overfitting or optimization challenges.
24. **What is the difference between an internal attention map and Grad-CAM?**
    Internal attention maps are learned weights computed directly during the forward pass, whereas Grad-CAM is a post-hoc gradient-weighted visualization technique.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Because it computes gradients backward from the classification score after the forward pass is complete to visualize feature importance.
26. **Does the attention model suppress irrelevant background regions?**
    Yes; spatial and self-attention modules actively reduce activation weights on background clutter (like soil and leaves).
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    Yes; correct predictions exhibit tight focus on the core object, whereas incorrect predictions often activate background or ambiguous regions.
28. **Which flower classes benefit most from attention?**
    Visually complex or structurally similar classes (e.g., roses vs. tulips) benefit most by resolving fine-grained inter-class ambiguities.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
    A confusion matrix reveals specific inter-class confusions, error distribution, and per-class performance biases.
30. **Why is macro F1 useful in addition to accuracy?**
    Macro F1 computes the unweighted average of class-wise F1 scores, preventing majority class bias from masking poor performance on minority classes.
31. **What is meant by the accuracy–complexity trade-off?**
    It describes the balance between maximizing classification accuracy and minimizing parameter count, memory footprint, and computational latency.
32. **What is meant by the F1–inference-time trade-off?**
    It evaluates whether performance gains (Macro F1) justify the increased per-image execution latency introduced by complex modules.
33. **Why must every ablation study use the same dataset split?**
    To ensure that performance variations stem strictly from architectural changes rather than data distribution shifts across subsets.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    To guarantee experimental fairness by controlling hyperparameters, learning schedules, and optimization environments.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    Attention models yield more compact object localization and superior background suppression than unaugmented baselines.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.

12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity}.\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

SOURCE CODE

https://github.com/sriwhatsnext/dllab/tree/main/Week00809

42. References

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